

RESULTS

# Results

01 · MANN-WHITNEY U

nonparametric two-group comparison

Table 1. Rank summary

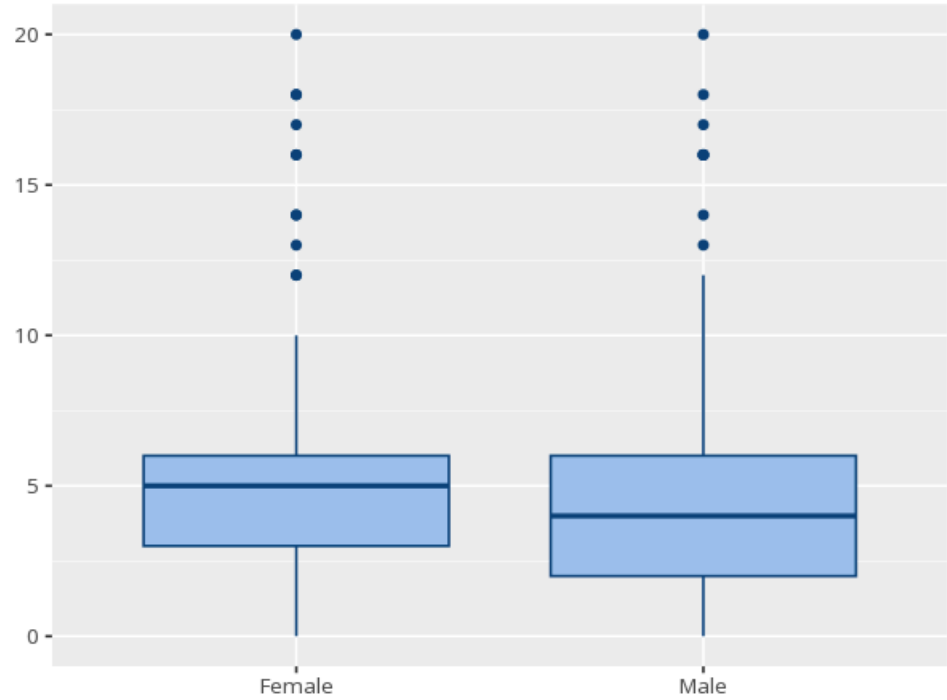
| Group  | N   | Mean rank | Median | IQR  | Sum of ranks |
|--------|-----|-----------|--------|------|--------------|
| Female | 120 | 124.30    | 5.00   | 3.00 | 14916.00     |
| Male   | 120 | 116.70    | 4.00   | 4.00 | 14004.00     |

Table 2. Mann-Whitney test

| U                                                             | Z    | p    | r [95% CI]         |
|---------------------------------------------------------------|------|------|--------------------|
| 7656                                                          | 0.85 | .394 | 0.06 [−0.08, 0.21] |
| Hodges-Lehmann median difference = 0.00, 95% CI [−0.00, 1.00] |      |      |                    |

r is the rank-biserial effect size. The Hodges-Lehmann estimate is the median of all between-group score differences (a location shift); its CI is from the same wilcox.test.

Figure. Distribution by group



## Understanding the numbers

**IQR.** The middle 50% spread of scores within each group (descriptive, not part of the test itself). Here, the IQR is 3.00 for Female and 4.00 for Male.

**Mean rank.** The average rank of each group's scores when all scores are ranked together - the quantity the test compares. Here, the mean rank is 124.30 for Female and 116.70 for Male.

**Median.** The middle value of each group's scores. Here, the median is 5.00 for Female and 4.00 for Male.

**N.** The number of observations in each group. Here, N = 120 for Female and 120 for Male.

**p.** The probability of a rank difference this large (or larger) if the two groups truly had the same distribution. Here,  $p = .394$  - at or above the conventional .05 threshold, no systematic difference detected.

**r.** The rank-biserial effect size for the difference between the two groups. Here,  $r = 0.06 [-0.08, 0.21]$ .

**Sum of ranks.** The total of the ranks assigned to each group's scores - the quantity U is calculated from. Here, the sum of ranks is 14916.00 for Female and 14004.00 for Male.

**U.** The Mann-Whitney test statistic, based on how the two groups' ranks compare. Here,  $U = 7656$ .

**Z.** The standardized test statistic used to compute p. Here,  $Z = 0.85$ .

**Hodges-Lehmann estimate.** The median of every possible between-group score difference - a robust estimate of the location shift between the two groups. Here, the Hodges-Lehmann estimate is 0.00, 95% CI  $[-0.00, 1.00]$ .

### How to read this test

The nonparametric counterpart to the independent t-test, comparing whole distributions via ranks rather than means. A p below alpha means one group's values tend to be systematically higher or lower (stochastic dominance) - this equals a difference in medians only when the two groups have similar distribution shapes. r gives the effect size. Your significance threshold ( $\alpha$ ) is 0.05. Method: Wilcoxon rank sum test with continuity correction.

**APA template:** A Mann-Whitney U test gave  $U=7656$ ,  $Z=0.85$ ,  $p = .394$ ,  $r=.06 [-.08, .21]$ .

**Statistical basis:** Mann-Whitney U test (Mann, H. B., Whitney, D. R. (1947). "On a test of whether one of two random variables is stochastically larger than the other." Annals of Mathematical Statistics, 18(1), 50-60.); Rank-biserial effect size r (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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